

OMKAR GANESH MAHABDI

AI/ML ENGINEER | GENERATIVE AI | LLM | RAG | PYTHON

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience in Python development, Artificial Intelligence, Machine Learning, Generative AI, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), semantic search, vector search, NLP workflows, and AI application development. Experienced in building production-oriented AI solutions using Python, PyTorch, FAISS, SentenceTransformers, Llama 2, FastAPI, Supabase, WebSockets, and LLM workflows. Developed AI-powered document search and retrieval systems that reduced document search latency from 45 seconds to under 500 milliseconds and improved semantic retrieval accuracy to 92%+. Strong experience in document ingestion, text processing, embeddings, vector databases, model fine-tuning, conversational AI, OCR fallback processing, real-time AI response streaming, and AI-driven experimentation.

CORE TECHNICAL SKILLS

- Python, C++, JavaScript, TypeScript
- Deep Learning & Generative AI
- Large Language Models (LLMs)
- NLP & Prompt Engineering
- Model Fine-Tuning & QLoRA
- Scikit-learn, TensorFlow, PyTorch
- Llama 2, Llama 3, CodeLlama-7B
- Text-to-SQL & LLM Applications
- LLM Evaluation & Red-Teaming
- Hallucination Detection
- LangChain & LlamaIndex
- Retrieval-Augmented Generation (RAG)
- SentenceTransformers & FAISS
- Vector Search & Semantic Search
- Vector Databases
- FastAPI, REST APIs, WebSockets
- Server-Sent Events (SSE)
- Conversational AI & Chatbots
- PDF Ingestion & Text Chunking
- OCR & Unstructured Data Extraction
- MySQL, PostgreSQL, DuckDB, MongoDB
- Query Optimization & CTEs
- Window & Ranking Functions
- Git, Docker, Supabase, Linux
- OpenCV & MediaPipe
- Netlify CI/CD

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – PYTHON, AI/ML & LLM WORKFLOWS

eInfochips (An Arrow Company)

January 2025 – Present

Hybrid, Pune, Maharashtra

- Develop and optimize AI-powered document search and retrieval solutions using Python, FAISS, SentenceTransformers, and Llama 2, reducing document search latency from 45 seconds to under 500 milliseconds.
- Improve semantic retrieval performance to 92%+ accuracy through optimized vector search, embedding improvements, and domain-specific model training based on systematic analysis of retrieval failure cases.
- Design and develop full-stack AI chat applications using Python, FastAPI, React, FAISS, Supabase, WebSockets, and LLM workflows for document-based question answering.
- Build end-to-end PDF ingestion and document processing workflows covering requirement extraction, text processing, semantic search, OCR fallback, and automated report generation.
- Implement real-time WebSocket streaming to deliver responsive AI-generated answers and support multi-turn conversational workflows.
- Analyze embedding quality degradation across different document categories and identify limitations of generic embeddings for domain-specific content.
- Retrain and fine-tune embedding models for domain-specific documents to improve retrieval consistency and accuracy across product categories.
- Integrate authentication and session management using Supabase for AI-powered applications.
- Develop automated PDF report generation with structured formatting, section organization, and summarized AI-generated content.
- Apply Python-based problem solving, machine learning concepts, vector databases, NLP workflows, and data-driven experimentation to improve AI application performance.

AI/ML PROJECTS

AI-POWERED CHATBOT

Python, FastAPI, FAISS, SentenceTransformers, Llama 2, REST API, Supabase, WebSockets/SSE

- Developed an AI-powered Retrieval-Augmented Generation (RAG) chatbot enabling users to upload PDF documents and interact with their content through natural-language questions.
- Engineered an end-to-end RAG pipeline covering document ingestion, text chunking, embedding generation, semantic retrieval, vector storage, and LLM-based response generation.
- Implemented SentenceTransformers-based embeddings and FAISS vector similarity search to achieve sub-500ms query latency.
- Integrated Llama 2 for context-aware response generation and optimized retrieval workflows to improve response relevance and efficiency.
- Added OCR fallback processing for low-quality scanned PDFs to improve document text extraction and downstream retrieval performance.

- Implemented WebSocket-based real-time response streaming and session management for multi-turn conversational context.
- Built automated PDF report generation with structured summaries, section headers, and formatted outputs.

LLM-POWERED SQL QUERY GENERATOR

Python, LLM, CodeLlama-7B, QLoRA, Text-to-SQL, FastAPI, SQL

- Developed a natural-language-to-SQL generation workflow that enables users to express database queries in plain English and receive executable SQL statements.
- Fine-tuned CodeLlama-7B on 80K text-to-SQL pairs using QLoRA, achieving 79% execution accuracy on the Spider benchmark.
- Designed the solution for API-based inference using FastAPI, supporting more than 500 queries per day with average response latency below 900ms.
- Applied LLM fine-tuning and text-to-SQL techniques to improve automated database query generation and reduce the need for manual SQL authoring.

LLM EVALUATION & RED-TEAMING FRAMEWORK

DeepEval, LangSmith, LLM Evaluation, Adversarial Testing

- Built an automated LLM evaluation framework to systematically identify hallucinations, bias, failure modes, and unsafe model behavior before deployment.
- Designed automated testing workflows covering 500+ adversarial prompts across six failure categories using DeepEval and LangSmith.
- Measured a 23% hallucination rate on edge-case prompts and used evaluation findings to identify targeted model improvement opportunities.
- Applied targeted fine-tuning based on evaluation results, reducing identified model violations by 61%.
- Structured repeatable evaluation and red-teaming workflows to improve LLM reliability and deployment readiness.

MULTIMODAL IMAGE + TEXT CLASSIFIER

CLIP, Computer Vision, NLP, Image Classification, Text Classification

- Built a multimodal classification pipeline combining image and product-title information to automatically tag product listings.
- Developed a CLIP-based image and text classification workflow covering 100K product listings and achieving 89% classification accuracy.
- Combined visual and textual representations to improve product categorization beyond keyword-based classification approaches.
- Reduced manual tagging effort by 64% compared with a keyword-only baseline.
- Implemented low-quality listing detection that enabled problematic listings to be flagged 4x faster.

CERTIFICATIONS

- OCI Certified Data Science Professional – Oracle
- Python Essentials 2 – Cisco
- Data Analysis with Python – freeCodeCamp
- Data Science Essentials With Python – Cisco
- Data Analytics Essentials – Cisco
- Artificial Intelligence Fundamentals – IBM
- OCI Generative AI Professional – Oracle
- Oracle AI Vector Search Professional – Oracle
- Deloitte Australia – Data Analytics Job Simulation
- Programming with Python 3.X – Skillup

ACHIEVEMENTS

- Ranked 872 among 445,000+ participants in the TCS CodeVita Global Coding Challenge, representing participation across 94 countries.
- Awarded a Merit Scholarship of INR 77,500 for exceptional academic performance.
- Finalist – HackMIT.
- Participant – Mercor Hackathon.

EDUCATION

BACHELOR OF TECHNOLOGY (BTECH) – COMPUTER SCIENCE ENGINEERING
October 2025

MIT World Peace University

August 2022 –

Pune, Maharashtra

CBSE CLASS XII – SCIENCE STREAM

Army Public School

August 2019 – August 2020

Pune, Maharashtra